Tutorium - 9

Gruppe 4B, 0337
Do, $10^{10} - 11^{45}$

T9.1: (a) $f: [a,b] \longrightarrow \mathbb{R}$ stetig differenzierbar

Ges: $\int_a^b f(x) f'(x)\, dx$

der Stammfunktion von $x \mapsto F'(f(x))\, f'(x)$
ist $x \mapsto F(x)$ (Kettenregel)
Hier ist $F'(y) = y$, also $F(y) = \frac{1}{2} y^2$

damit ist $\frac{1}{2} f^2(x)$ der Stamfkt. von $f(x) f'(x)$

und:

$$\int_a^b f(x)\, f'(x)\, dx = \left[\frac{1}{2}(f(x))^2\right]_a^b = \frac{1}{2}\left[(f(b))^2 - (f(a))^2\right]$$

(b) sei $g(x) = cx + d$, $g'(c) = c$. Dann:

$$\int_a^b f(cx+d)\, dx = \int_a^b f(g(x))\, dx = \int_a^b f(g(x)) \frac{g'(x)}{c}\, dx$$

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$$\overset{g(x)=y}{=} \frac{1}{c} \int_{g(a)}^{g(b)} f(y)\, dy = \frac{1}{c}\left(F(g(b)) - F(g(a))\right)$$

$$= \frac{1}{c}\left[F(cb+d) - F(ca+d)\right]$$

(c) $\int \frac{\ln x}{x}\, dx$; $f(x) = \ln(x)$, $f'(x) = \frac{1}{x}$

- $\int \frac{\ln x}{x}\,dx = \int f(x)\,f'(x)\,dx \overset{(a)}{=} \frac{1}{2}(\ln x)^2$
- $\int \sqrt{3x+4}\;dx$, (b) mit $c=3$, $d=4$, $f(x)=\sqrt{x}$

$$\int \sqrt{3x+4}\;dx = \frac{1}{3}\,\frac{2}{3}(3x+4)^{3/2} = \frac{2}{9}\,(3x+4)^{3/2}$$

T9.2:

(a) $x \mapsto \frac{1}{\cos^2(x)}$, aus $\tan'(x) = \frac{1}{\cos^2(x)}$ folgt

$$\int \frac{1}{(\cos(x))^2}\,dx = \tan(x) \quad \text{auf } (k\pi - \pi/2,\ k\pi + \pi/2) \quad k \in \mathbb{Z}.$$

(b) $x \mapsto \tan(x)$, für $X \in (-\frac{\pi}{2}, \pi/2)$:

$$\int_0^X \tan(x)\,dx = \int_0^X \frac{\sin(x)}{\cos(x)}\,dx = \int_0^X \frac{-\cos'(x)}{\cos(x)}\,dx$$

$$= -\int_{\cos(0)}^{\cos(X)} \frac{1}{y}\,dy = -\ln(\cos(X))$$

aus $\tan(x+\pi) = \tan x \;\wedge\; \cos(x+\pi) = -\cos(x)$

gilt $\int \tan(x)\,dx = -\ln|\cos x|$ auf $(k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2})$ $k \in \mathbb{Z}.$

(c) $\arctan'(x) = \frac{1}{1+x^2}$ und damit:

$$\int \frac{dx}{1+x^2} = \arctan(x)$$

(d) $$\int_0^X \arctan x \, dx = \int_0^X 1 \cdot \arctan x \, dx$$

$$= x \arctan(x)\Big|_0^X - \int_0^X \frac{dx}{1+x^2} \cdot x$$

$$= X \arctan(X) - \frac{1}{2} \int_0^X \frac{d(1+x^2)}{1+x^2}$$

$$= X \arctan(X) - \frac{1}{2} \ln|1+X^2|$$

Also ist:

$$\int \arctan x \, dx = x \arctan x - \frac{1}{2} \ln|1+x^2| + C$$

(e) $$\overbrace{\int_0^X \sin^2 x \, dx}^{=:I} = -\int_0^X \sin x \, d(\cos x)$$

$$\overset{\text{P.I.}}{=} - \cos x \sin x\Big|_0^X + \int_0^X \cos x \, d(\sin x)$$

$$= - \cos X \cdot \sin X + \int_0^X \cos^2 x \, dx$$

$$= - \cos X \sin X + \int_0^X 1 - \sin^2 x \, dx$$

$$I = - \cos X \sin X + X - 0 - I$$

$$\Rightarrow \quad 2I = - \cos X \sin X + X \quad \Rightarrow \quad I = \frac{1}{2}(x - \cos x \sin x) + c.$$

T 9.3:

z.z.: $\int_0^1 x^n (1-x)^m \, dx = \dfrac{n!\, m!}{(n+m+1)!}$; $n, m \in \mathbb{N}_0$

• Wir beweisen dies mittels vollständige induktion nach n!

→ Induktionsanfang: $\int_0^1 x^n (1-x)^m \, dx = \dfrac{n!\, m!}{(n+m+1)!}$, $n = 0$

• $\int_0^1 x^0 (1-x)^m \, dx = \int_0^1 (1-x)^m \, dx = \underbrace{-\dfrac{(1-x)^{m+1}}{m+1}}_{\text{Stammfunktion von } (1-x)^m!} \Big|_0^1 = \dfrac{1}{m+1}$

• $\dfrac{m! \cdot \cancel{0!}\, 1}{\underbrace{(m+1)!}_{m!\,(m+1)}} = \dfrac{1}{m+1}$! ✓

→ Induktionsvoraussetzung:

$$\int_0^1 x^\nu (1-x)^m \, dx = \frac{\nu!\, m!}{(\nu+m+1)!} \quad \forall\, \nu \le n,\ m \in \mathbb{N}_0$$

→ Induktionsschritt: $n \rightsquigarrow n+1$

$$\int_0^1 x^{n+1} (1-x)^m \, dx = \frac{-1}{m+1} \int_0^1 x^{n+1} \, d\left((1-x)^{m+1}\right)$$

P.I.

$$= \underbrace{\left[-\frac{1}{m+1} \cdot x^{n+1} (1-x)^{m+1}\right]_0^1}_{✓} + \frac{n+1}{m+1} \int_0^1 (1-x)^{m+1} \cdot x^n \, dx$$

$$= 0 + \frac{n+1}{\cancel{m+1}} \cdot \frac{n!\,\cancel{(m+1)!}^{\,m!\,\cancel{(m+1)}}}{(n+(m+1)+1)!} = \frac{(n+1)!\ m!}{((n+1)+m+1)!} \quad \square$$